

Pranav Acharya

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🔗 Pranav-Acharya1

Profile

Accomplished **Electrical Engineering PhD** and **Masters in Theoretical Physics**, with experience in **quantum transport modelling** and **device physics**. Published **4 first author papers** [1–4] leveraging **Non-Equilibrium Green's Function (NEGF)** to quantify the impact of device variation on **Resonant Tunnelling Diodes (RTDs)**. This variation was shown to be able to encode information, hence RTDs can compose **Physical Unclonable Functions (PUFs)**. Seeking a **Research Scientist** role to apply physics theory in the development of **next-generation quantum and nano-electronic devices**.

Skills

Nanoelectronic Device Modelling

- Modelled impact of device variation on quantum tunnelling based devices (RTDs) using Non-Equilibrium Green's Function (NEGF) formalism, delivering **4 first-author research projects** [1–4]
- Simulated excitonic electron-hole complexes in quantum dots using Quantum Monte Carlo methods
- **Diverse modelling expertise**, including **RTDs** [1–4], **nanosheet transistors** [5, 6], and **quantum dots**

Computational Physics and Programming

- Extended the device variation capability [1–4] of custom **C++ TCAD software** [🔗](#), using **Git** for version control, and utilised **Python** (NumPy, SciPy, Matplotlib) and MATLAB for numerical analysis and data post-processing
- Collaborated to accelerate simulations **by 60%** using a **Diffusion ML model** (SISPAD conference paper) [5]
- Automated simulations on Linux HPC environments using Python scripts and queue managers (Slurm/SGE)

Collaboration, Communication, and Comprehension

- Co-authored 6 peer-reviewed papers [1–6], collaborating across international institutions
- Presented research at **EuroSOI-ULIS 2024** [1] and wrote a [blog](#) [🔗](#), as well as a [tutorial](#) [🔗](#) on quantum ML
- Actively upskill in computational advances: [Bath ML](#) [🔗](#) & [Qiskit](#) [🔗](#) Summer Schools and [UK Quantum Showcase](#) [🔗](#)

Education

PhD, Electrical and Electronic Engineering, **University of Glasgow**

2021–2025

- [Simulation of Resonant Tunnelling Diodes with the Non-Equilibrium Green's Function Formalism](#) [🔗](#)

MPhys, Theoretical Physics, **First class honours**, **Lancaster University**

2017–2021

- Thesis project: Quantum Monte Carlo simulations of 3D type 2 Quantum Dots

Publications

- [1] Pranav Acharya and Vihar Georgiev. “Interface roughness in resonant tunnelling diodes for physically unclonable functions”. In: *Solid-State Electronics* (2025), p. 109131.
- [2] Pranav Acharya et al. “Analysis of Random Discrete Dopants Embedded Nanowire Resonant Tunnelling Diodes for Generation of Physically Unclonable Functions”. In: *IEEE Transactions on Nanotechnology* 23 (2024), pp. 815–821.
- [3] Pranav Acharya et al. “Impact of interface roughness correlation on resonant tunnelling diode variation”. In: *Scientific Reports* 15.1 (2025), p. 26815.
- [4] Pranav Acharya et al. “Sensitivity of resonant tunneling diodes to barrier variation and quantum well variation: A NEGF study”. In: *Micro and Nanostructures* 207 (2025), p. 208264.
- [5] P. Aleksandrov, P. Acharya, and V. Georgiev. “Diffusion-Based Machine Learning Method for Accelerating Quantum Transport Simulations in Nanowire Transistors”. In: *2024 International Conference on Simulation of Semiconductor Processes and Devices (SISPAD)*. 2024, pp. 1–4.
- [6] Tongfei Liu et al. “The study of electron mobility on ultra-scaled silicon nanosheet FET”. In: *Physica Scripta* 99.7 (2024), p. 075410.